

This presentation can be downloaded here:

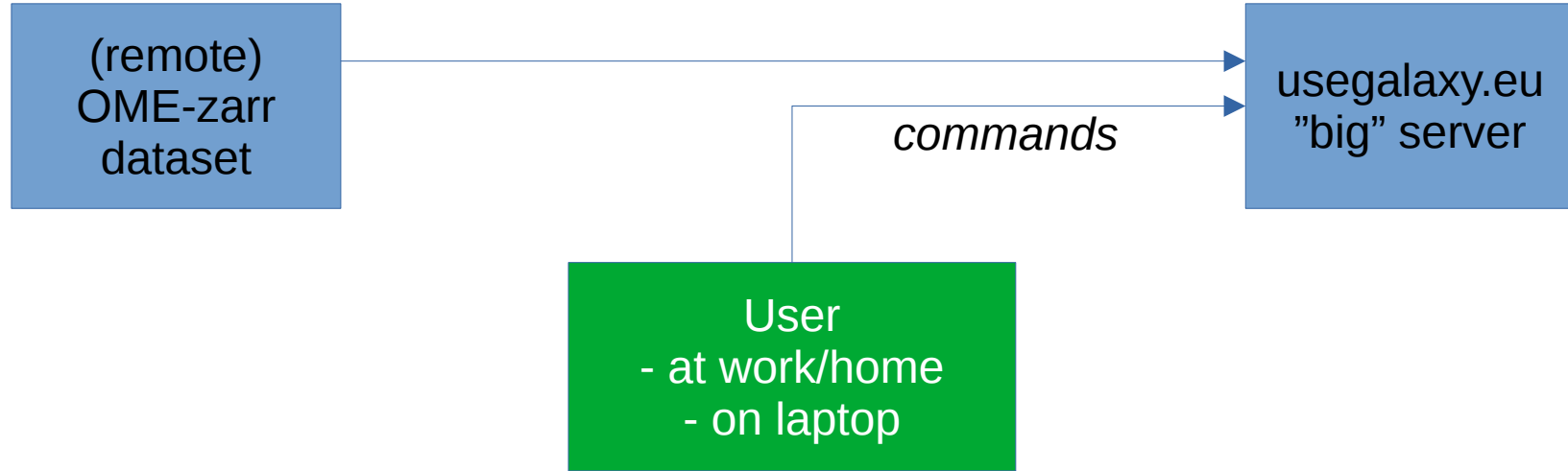
https://github.com/xulman/tools-iuc/tree/trackastra_fixes/tools/trackastra/presentation

The Goal

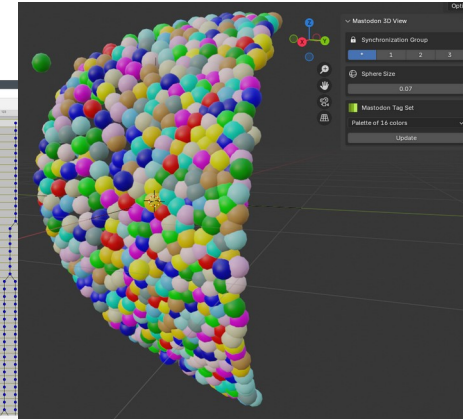
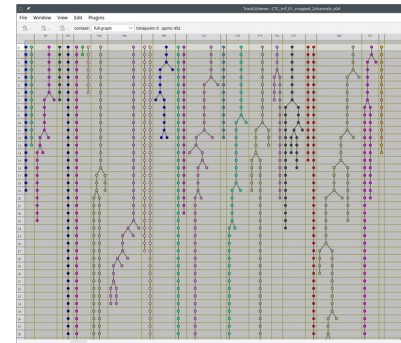
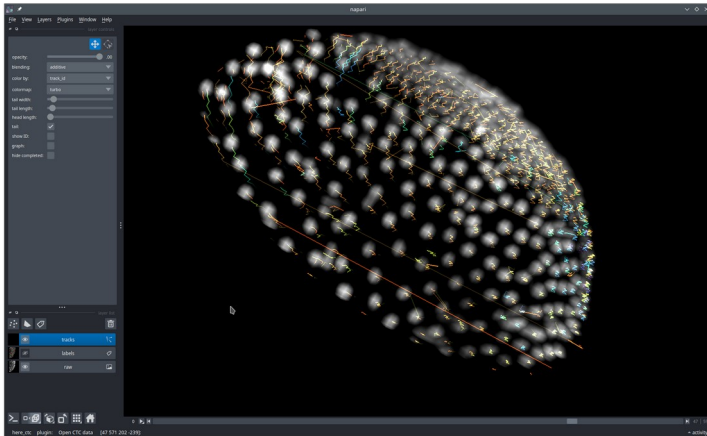
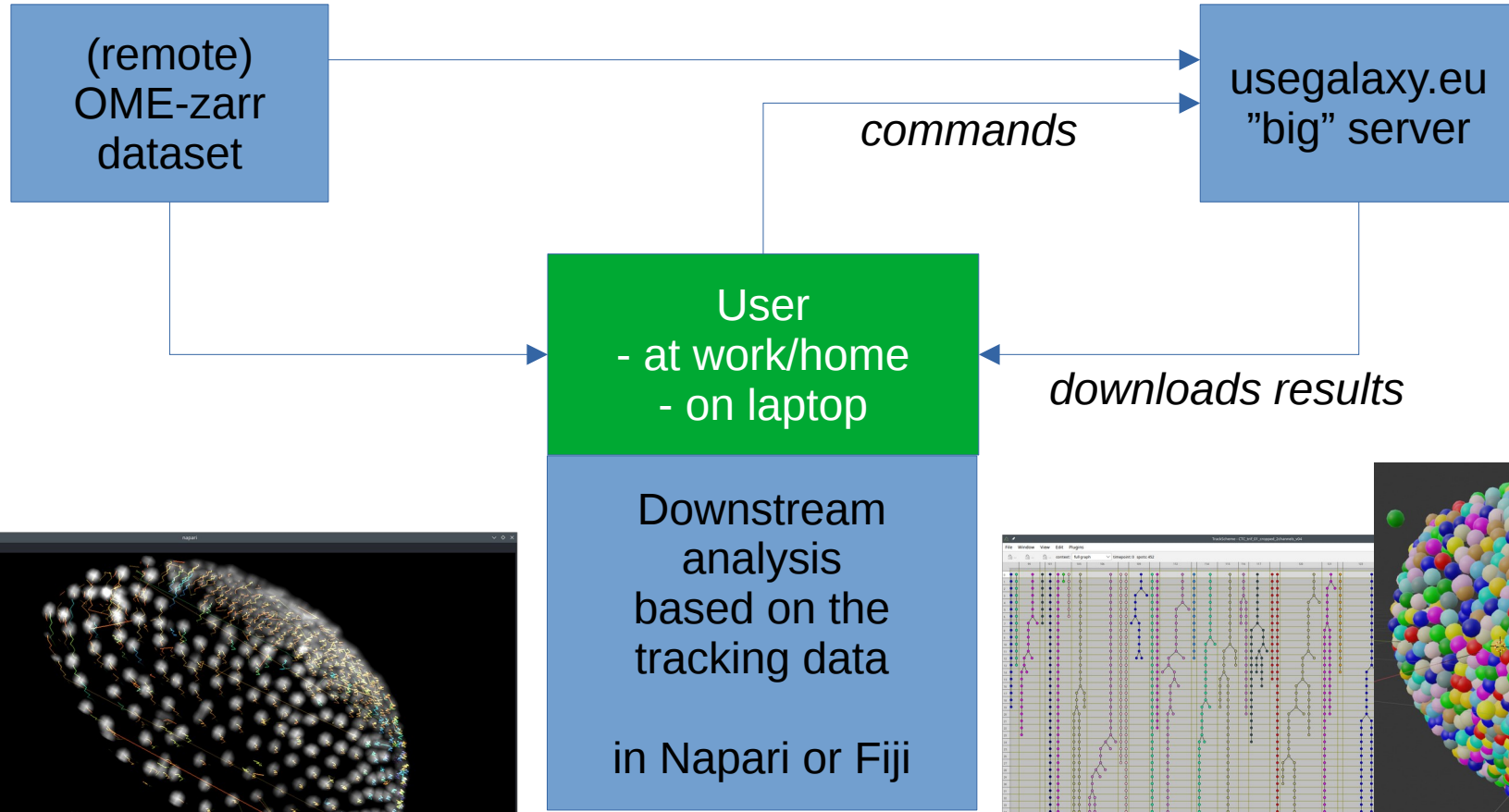
To show a tracking pipeline on OME-zarr image data.

- Pipeline is open source & license, freely accessible, in Python (nearly FAIR :)
https://github.com/xulman/tools-iuc/tree/trackastra_fixes/tools/trackastra
- Based on the popular (one-click) Cellpose3 and Trackastra
- Credits to the original Authors: Carsen Stringer, Marius Pachitariu, Martin Weigert
- Status as of 18th March 2026:
 - Aspirant for a proper Galaxy node
 - Whatever you learn today, it applies to the future node
 - ... It is perhaps a little resources hungry...
 - Still usable?
 - Yeah, you can run in free tier of Galaxy!
 - That's why we are here today :)

The Grand Scheme of Things



The Grand Scheme of Things



Trackastra

- AI-including cell (and nuclei) tracker for 2D+t and 3D+t
- Homepage: <https://github.com/weigertlab/trackastra>
- Inputs:
 - Mask images – to show subjects of tracking
 - Raw images – to help tracker find pairs
 - *Loaded fully as numpy data*
- Outputs:
 - Napari tracks (object as such), directly inside Napari
 - Graph Exchange File Format (GEFF) <https://github.com/live-image-tracking-tools/geff>
 - Cell Tracking Challenge <https://celltrackingchallenge.net/>
 - *GEFF & CTC formats hold masks internally*

Cellpose v3

- AI-based cell (and nuclei) segmenter for 2D and 3D
- Homepage: <https://github.com/MouseLand/cellpose>
- Using v3 (prior v4 – SAM-enabled), it is feasible on CPUs (no GPUs)
- Available in this pipeline for cases when only raw images are provided
 - On-the-fly segmentation for Trackastra
 - Results not stored aside (*TODO: Should be*)
 - In our hands: Needs objects diameter information
 - *It can be compute heavy*
 - *Considered for emergency cases*
 - *Better to segment properly outside this*

Controls of this Pipeline

- Input URL
- Input resolution level
- Raw images channel
- Mask images channel
- Output folder
- Tracking Options → → → →
 - Tracking Options object
 - downscale_factor_x: 1.0
 - downscale_factor_y: 1.0
 - downscale_factor_z: 1.0
 - start_from_tp: 0
 - end_at_tp: -1
 - segmentation_model: 'cyto3' or 'nuclei'
 - objects_diameter_px : 25
 - tracking_model: 'ctc'

These all will be parameters of the future Galaxy node.

Controls of this Pipeline

- **Input resolution level**

Why downscaling?

- Trackastra reads the full input.
- Spatial extent is good to know, but less important...

- Tracking Options object

- **downscale_factor_x: 1.0**
- **downscale_factor_y: 1.0**
- **downscale_factor_z: 1.0**
- **start_from_tp: 0**
- **end_at_tp: -1**
- **segmentation_model: 'cyto3' or 'nuclei'**
- **objects_diameter_px : 25**
- **tracking_model: 'ctc'**



Geometry of OME-zarr

OME NGFF Validator

Image: CTC_trif_01_cropped_2channels_v04.zarr

Validating: [/CTC_trif_01_cropped_2channels_v04.zarr/zattrs](#)
Using version 0.4. Using schema: [/v0.4/schemas/image.schema](#)

Open with:   

```
{
  "multiscales": [
    {
      "datasets": [...], // 4 items
      "name": "/",
      "axes": [
        {
          "name": "t",
          "type": "time"
        },
        {
          "name": "c",
          "type": "channel"
        },
        {
          "name": "z",
          "type": "space"
        },
        {
          "name": "y",
          "type": "space"
        },
        {
          "name": "x",
          "type": "space"
        }
      ],
      "version": "0.4"
    }
  ]
}
```

Multiscale 0
4 Datasets checked ✓

Path [s0/zarray](#)

	Array	Chunk
Bytes	31.88 GB	524.29 KB
Shape	[60,2,837,522,304]	[1,1,64,64,64]
Counts	60,2,14,9,5	75600 Chunks
Type	<u2	numpy.ndarray

Load chunk: ☐ 0 ☐ 0 ☐ 0 ☐ 0 ☐ 0

► s0/zarray

Path [s1/zarray](#)

	Array	Chunk
Bytes	3.98 GB	524.29 KB
Shape	[60,2,418,261,152]	[1,1,64,64,64]
Counts	60,2,7,5,3	12600 Chunks
Type	<u2	numpy.ndarray

Load chunk: ☐ 0 ☐ 0 ☐ 0 ☐ 0 ☐ 0

► s1/zarray

OME NGFF Validator

Image: CTC_CHOas2D_01_2channels_v04.zarr

Validating: [/CTC_CHOas2D_01_2channels_v04.zarr/zattrs](#)
Using version 0.4. Using schema: [/v0.4/schemas/image.schema](#)

Open with:   

```
{
  "multiscales": [
    {
      "datasets": [...], // 4 items
      "name": "/",
      "axes": [
        {
          "name": "t",
          "type": "time"
        },
        {
          "name": "c",
          "type": "channel"
        },
        {
          "name": "y",
          "type": "space"
        },
        {
          "name": "x",
          "type": "space"
        }
      ],
      "version": "0.4"
    }
  ]
}
```

Multiscale 0
4 Datasets checked ✓

Path [s0/zarray](#)

	Array	Chunk
Bytes	41.73 MB	4.10 KB
Shape	[92,2,443,512]	[1,1,64,64]
Counts	92,2,7,8	10304 Chunks
Type	lu1	numpy.ndarray

Load chunk: ☐ 0 ☐ 0 ☐ 0 ☐ 0 ☐ 0

► s0/zarray

Path [s1/zarray](#)

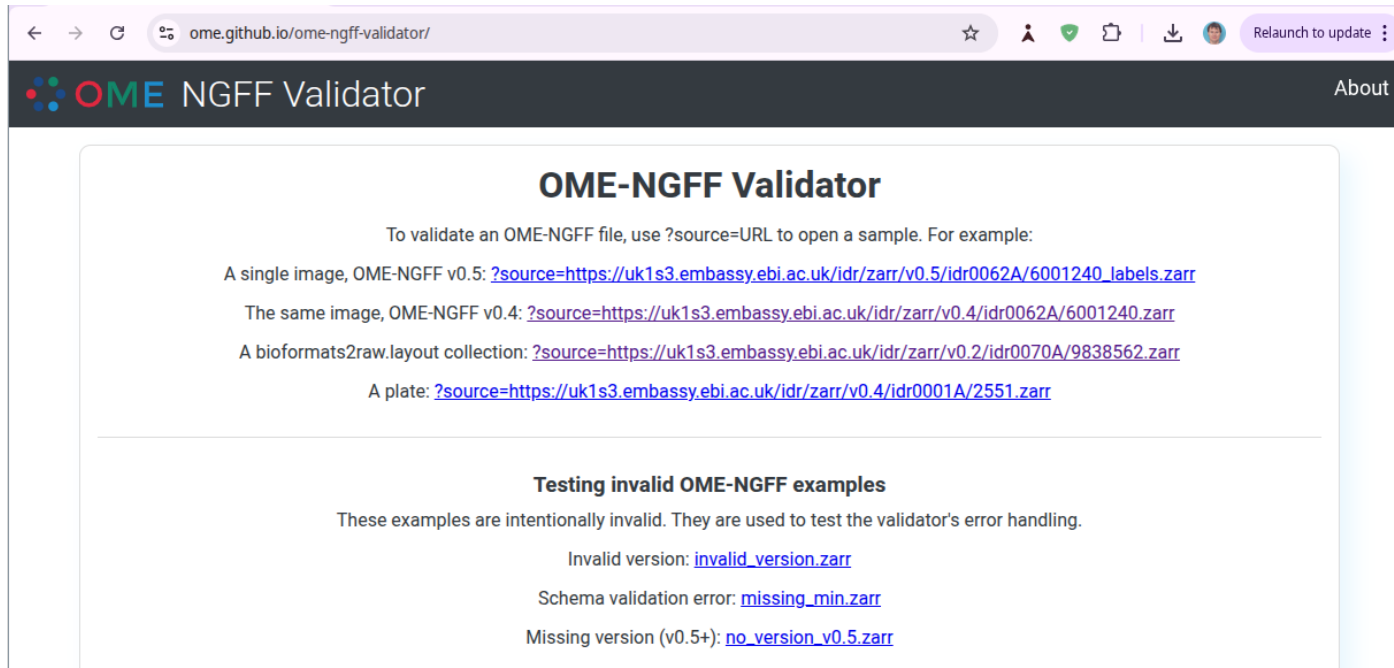
	Array	Chunk
Bytes	10.46 MB	4.10 KB
Shape	[92,2,222,256]	[1,1,64,64]
Counts	92,2,4,4	2944 Chunks
Type	lu1	numpy.ndarray

Load chunk: ☐ 0 ☐ 0 ☐ 0 ☐ 0 ☐ 0

► s1/zarray

Geometry of OME-zarr

- `pixi init; pixi add ome-zarr; pixi run ome_zarr view /path/to/dataset.zarr`
...turns local path to “network” URL
- `https://ome.github.io/ome-ngff-validator/?source=YOUR_URL`



On Galaxy via Interactive Jupyter (Notebook)

usegalaxy.eu/interactivetool/ep/n16cm1qtltn8/f0xqueim7cs/python/lab/workspaces/auto-a/tree/Untitled.ipynb

File Edit View Run Kernel Tabs Settings Help

Untitled.ipynb

Python 3 (ipykernel)

```
[ ]: !pip install trackastra 'cellpose<4.0'
```

```
[7]: !pip list | grep cellpose
!pip list | grep trackastra

cellpose          3.1.1.3
trackastra         0.5.2
```

```
[5]: import torch
torch.cuda.is_available()
```

```
[5]: False
```

```
[ ]: !wget https://raw.githubusercontent.com/xulman/tools-iuc/refs/heads/trackastra_fixes/tools/trackastra/trackastra_wrapper.py
```

```
[14]: import trackastra_wrapper as TAW
tops = TAW.default_tracking_options.copy()
tops['end_at_tp']=3
dataset_url = 'https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_CHO0as2D_02_2channels_v04.zarr'
dataset_scale = 2
select_raw = [0]
select_masks = [1]
```

```
[15]: tops
```

```
[15]: {'downscale_factor_x': 1.0,
'downscale_factor_y': 1.0,
'downscale_factor_z': 1.0,
'start_from_tp': 0,
'end_at_tp': 3,
'segmentation_model': 'cyto3',
'tracking_model': 'ctc'}
```

```
[16]: t = TAW.track_entry(dataset_url,dataset_scale, select_raw,select_masks, 'result_CHO02_as_ctc_fromJntb2', tops)
```

provided input data: 92 images of shape (1, 111, 128) pixels
resizing, going to scale images: False
preparing new input data: 4 images of shape [1, 111, 128] pixels
memory allocation for segmentation results started...
memory allocation for raw images started...
resizing started...
done resizing frame 0, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 1, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 2, target image size was (1, 111, 128) (source size was (1, 111, 128))
INFO:trackastra.model.model:Loading model state from /home/jovyan/.local/share/trackastra/models/ctc/model.pt
done resizing frame 3, target image size was (1, 111, 128) (source size was (1, 111, 128))
resizing done
/home/jovyan/.local/share/trackastra/models/ctc already downloaded, skipping.
INFO:trackastra.model.model_api:Using device cpu
INFO:trackastra.model.model_api:Default batch size = 1 for model on cpu.
INFO:trackastra.model.model_api:Predicting weights for candidate graph

Simple 2 1 Python 3 (ipykernel) | Idle Mode: Command Ln 1, Col 39 Untitled.ipynb

On Galaxy via Interactive Jupyter (Terminal)

usegalaxy.eu/interactivetool/ep/n16cm1qtltn8/lf0xqueim7cs/python/lab/workspaces/auto-a

File Edit View Run Kernel Tabs Settings Help

+

galaxy_inputs 13h ago

outputs 13h ago

result_CHO02_as_... 11h ago

result_CHO02_as_... 2h ago

result_CHO02_as_... 39m ago

pixi 12h ago

pixi_reference.lock 12h ago

pixi-x86_64-unkno... 20h ago

pixi.lock 12h ago

pixi.toml 12h ago

result.zip 1h ago

trackastra_wrapp... 1h ago

Untitled.ipynb 5m ago

```
(base) jovyan@a2c1f42a74e9:/data/jwd87/pulsar_staging/98431919/working/jupyter$ wget -q https://github.com/prefix-dev/pixi/releases/download/v0.66.0/pixi-x86_64-unknown-linux-musl
(base) jovyan@a2c1f42a74e9:/data/jwd87/pulsar_staging/98431919/working/jupyter$ chmod u+x pixi-x86_64-unknown-linux-musl
(base) jovyan@a2c1f42a74e9:/data/jwd87/pulsar_staging/98431919/working/jupyter$ ./pixi-x86_64-unknown-linux-musl run --frozen python
Python 3.14.3 | packaged by conda-forge | (main, Feb 9 2026, 21:56:02) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import trackastra_wrapper as TAW
... tops = TAW.default_tracking_options.copy()
... tops['end_at_tp'] = 20
... dataset_url = 'https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_CH0as20_02_2channels_v04.zarr'
... dataset_scale = 2
... select_raw = [0]
... select_masks = [1]
...
>>> tops
{'downscale_factor_x': 1.0, 'downscale_factor_y': 1.0, 'downscale_factor_z': 1.0, 'start_from_tp': 0, 'end_at_tp': 20, 'segmentation_model': 'cyto3', 'objects_diameter_px': 25, 'tracking_model': 'ctc'}
>>> t = TAW.track_entry(dataset_url, dataset_scale, select_raw, select_masks, 'result_CHO02_as_ctc2', tops)
version mismatch: detected: FormatV04, requested: FormatV05
version mismatch: detected: FormatV04, requested: FormatV05
provided input data: 92 images of shape (1, 111, 128) pixels
resizing, going to scale images: False
preparing new input data: 21 images of shape [1, 111, 128] pixels
memory allocation for segmentation results started...
memory allocation for raw images started...
resizing started...
done resizing frame 0, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 1, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 2, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 3, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 4, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 5, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 6, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 7, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 8, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 9, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 10, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 11, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 12, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 13, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 14, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 15, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 16, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 17, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 18, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 19, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 20, target image size was (1, 111, 128) (source size was (1, 111, 128))
resizing done
/home/jovyan/.local/share/trackastra/models/ctc already downloaded, skipping.
INFO:trackastra.model.model:Loading model state from /home/jovyan/.local/share/trackastra/models/ctc/model.pt
```

On Linux Terminal (of a Strong Workstation)

```
xulman@login3 ~
$ SLURM_FTA_getOneCPUOneNode_INTERACTIVE.cml
salloc: Pending job allocation 4045523
salloc: job 4045523 queued and waiting for resources
salloc: job 4045523 has been allocated resources
salloc: Granted job allocation 4045523
salloc: Waiting for resource configuration
salloc: Nodes cn096 are ready for job
xulman@cn096 ~
$ cd /mnt/proj2/open-28-13/Galaxy/galaxy_trackastra_wrapper_REPO/tools/trackastra
xulman@cn096 /mnt/proj2/open-28-13/Galaxy/galaxy_trackastra_wrapper_REPO/tools/trackastra
$ pipx run python
python 3.14.5 | packaged by conda-forge | (main, Feb 9 2026, 21:56:02) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> import trackastra_wrapper as TAW
... tops = TAW.default_tracking_options.copy()
... dataset_url = 'https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_CH0as2D_02_2channels_v04.zarr'
... dataset_scale = 2
... select_raw = [0]
... select_masks = [1]
>>> tops
{'downscale_factor_x': 1.0, 'downscale_factor_y': 1.0, 'downscale_factor_z': 1.0, 'start_from_tp': 0, 'end_at_tp': -1, 'segmentation_model': 'cyto3', 'tracking_model': 'ctc'}
>>> t = TAW.track_entry(dataset_url, dataset_scale, select_raw, select_masks, 'result_CH002_as_ctc2', tops)
version mismatch: detected: FormatV04, requested: FormatV05
version mismatch: detected: FormatV04, requested: FormatV05
provided input data: 92 images of shape (1, 111, 128) pixels
resizing, going to scale images: false
preparing new input data: 92 images of shape [1, 111, 128] pixels
memory allocation for segmentation results started...
memory allocation for raw images started...
resizing started...
done resizing frame 0, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 1, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 2, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 3, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 4, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 86, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 87, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 88, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 89, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 90, target image size was (1, 111, 128) (source size was (1, 111, 128))
done resizing frame 91, target image size was (1, 111, 128) (source size was (1, 111, 128))
resizing done
/home/xulman/.local/share/trackastra/models/ctc already downloaded, skipping.
INFO:trackastra.model.model:Loading model state from /home/xulman/.local/share/trackastra/models/ctc/model.pt
INFO:trackastra.model.model_api:Using device cpu
INFO:trackastra.model.model_api:Default batch size = 1 for model on cpu.
tracking started...
INFO:trackastra.model.model_api:Predicting weights for candidate graph
INFO:trackastra.data.wrfeat:Extracting features from 92 frames.
INFO:trackastra.data.wrfeat:Using single process for feature extraction
Extracting features: 100% | 92/92 [00:02:00]
:00, 32.261t/s]
INFO:trackastra.model.model_api:Building windows
Building windows: 100% | 89/89 [00:00:00]
0, 8500.351t/s]
INFO:trackastra.model.model_api:Predicting windows with batch size 1
Computing associations: 100% | 89/89 [00:07:00]
:00, 12.061t/s]
INFO:trackastra.model.model_api:Running greedy tracker
INFO:trackastra.tracking.tracking:Build candidate graph with delta_t=1
INFO:trackastra.tracking.tracking:Added 1196 vertices, 1177 edges
INFO:trackastra.tracking.tracking:Running greedy tracker
Greedly matched edges: 98% | 1155/1177 [00:00:00]
118009.821t/s]
tracking done
Converting graph to CTC results: 100% | 47/47 [00:00:00]
0, 2291.781t/s]
Saving masks: 100% | 92/92 [00:00:00]
00, 266.841t/s]
>>>
```

When tracking is done...

- ...I suggest to download and inspect locally
 - Tracking data is (usually) not super large
 - Desktop apps more powerfull than “in-browsering”

Napari available
as interactive
node on Galaxy

For Napari:

- `pip install napari pyqt napari-ome-zarr napari-ctc-opener`
- File → New img from clipboard (Ctrl+N)
- Console:

```
viewer.open('URL_to/dataset.zarr',  
plugin='napari-omē-zarr')
```

```
viewer.open('/path_to/result_as_ctc',  
plugin='napari-open-ctc')
```

For Fiji:

- Enable update sites:
Mastodon (and/or Trackmate),
CellTrackingChallenge
- Mastodon launcher, OME-NGFF,
Create .mastodon project, Open it
- Plugins → CTC → Import...
- View → Coloring → By track

Free Demo Datasets

- https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_CHOas2D_01_2channels_v04.zarr
- https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_CHOas2D_02_2channels_v04.zarr
- https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_huh7_01_2channels_v04.zarr
- https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_huh7_02_2channels_v04.zarr
- https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_trif_01_cropped_2channels_v04.zarr
- https://s3.cl2.du.cesnet.cz/35b9fef6_a5c7_4724_b7ad_0db97899a356:public/CTC_trif_02_cropped_2channels_v04.zarr